UNIVERSITY OF MORATUWA
Faculty of Engineering
Department of Electronic and Telecommunication Engineering
B.Sc.Engineering
Semester 5 (20 Batch) Examination

BM3122 MEDICAL IMAGING

Time allowed: *Two* hours November 2023

ADDITIONAL MATERIAL:

- An information sheet on common constants and physical properties is attached.

INSTRUCTIONS TO CANDIDATES:

- This paper contains 4 questions on 6 pages.
- This examination accounts for 50% of the module assessment. The total maximum mark attainable is 100. The marks assigned for each question & sections there of are indicated in square brackets.
- This is a closed book examination.
- Derivations are not required if they are not explicitly requested for in the question.
- Assume reasonable values for any data not given in or with the examination paper. Clearly state such assumptions.
- If you have any doubt as to the interpretation of the wording of a question, make your own decision, and clearly state it.
- All questions carry equal marks.
- Answer **ALL** questions.

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Information Sheet

Avogadro's number $= 6.02 \times 10^{23}$
Boltzmann's constant: $k = 1.38 \times 10^{-23}$ J/K
Density of water: $\rho_{\text{water}} = 1,000$ kg/m^3
Gyro-magnetic ratio of protons: $\gamma = 267.52 \times 10^6$ rad s^{-1}T^{-1}
Plank's constant: $h = 6.63 \times 10^{-34}$ Js

Some basic relationships – *Symbols have their usual meanings*

De Broglie's relationship:	$\Delta E = hf$
For spins in a strong magnetic field:	$N_{\text{parallel}} - N_{\text{anti-parallel}} = N_{\text{total}} \dfrac{\gamma h B_0}{4\pi k T}$
Net magnetization for spins in a strong magnetic field:	$\mathbf{M}_0 = \dfrac{\gamma h}{4\pi} \left(N_{\text{parallel}} - N_{\text{anti-parallel}}\right)$
The slice thickness of an MRI scan:	$T = \frac{2\Delta\omega_s}{\gamma G_{\text{slice}}}$
Image intensity of MRI pixel acquired using a Gradient Echo sequence:	$I(x,y) \propto \rho(x,y) \dfrac{\left(1 - e^{-\text{TR}/\text{T}_1}\right) \sin\alpha}{1 - e^{-\text{TR}/\text{T}_1} \cos\alpha} e^{-\frac{\text{TE}}{\text{T}_2^*}}$
Optimized tip angle for a given TR for maximum SNR:	$\alpha_{\text{Ernst}} = \cos^{-1} e^{-\frac{\text{TR}}{\text{T}_1}}$
The intensity of an axial MRI slice acquired using Spin Echo sequences:	$I(x,y) \propto \rho(x,y) \left(1 - e^{-\frac{\text{TR}}{\text{T}_1}}\right) e^{-\frac{\text{TE}}{\text{T}_2^*}}$
Radioactivity of a radioactive isotope:	$Q = \dfrac{dN}{dt} = \lambda N$
The number of radioactive nuclei at time, t:	$N(t) = N(t=0)e^{-\lambda t}$
The effective radiotracer half-life:	$\tau_{1/2,\text{eff}} = \dfrac{\tau_{1/2} \cdot \tau_{1/2,\text{bio}}}{\tau_{1/2} + \tau_{1/2,\text{bio}}}$

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Question 1.

(a) Different imaging modalities are used to view the interior of a body in order to diagnose, monitor, or treat medical conditions.

i. Name **three** imaging modalities that use ionizing radiation and **three** non ionizing modalities. [3]

ii. What are the parameters represented by pixels of each imaging modality in (a)i.? [4]

(b) An analogue-to-digital converter (ADC) takes an analogue input voltage from the medical imaging system and produces a digital output.

i. Name **three** important specifications of an ADC in an imaging system. [3]

ii. What is Quantization Error? [2]

iii. Propose **two** methods to minimize the relative contribution of the quantization error. [3]

(c) A small amount of a radioactive material called a radiotracer is injected to the patient in nuclear medicine imaging.

i. How much ^{113m}In activity should be purchased by Thursday at 4.00 pm to provide 300 kBq for an imaging experiment planned to conduct on Friday at 1.00 pm? The half-life of ^{113m}In is 1.7 hours. [5]

ii. How much time should elapse before a 100 GBq sample of ^{113m}In and a 20 GBq sample of ^{99m}Tc possess equal activities? The half-life of ^{99m}Tc is 6.0 hours. [5]

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Question 1

(a) i) ionizing - projected Xray, CT, Nuclear Medicine imaging

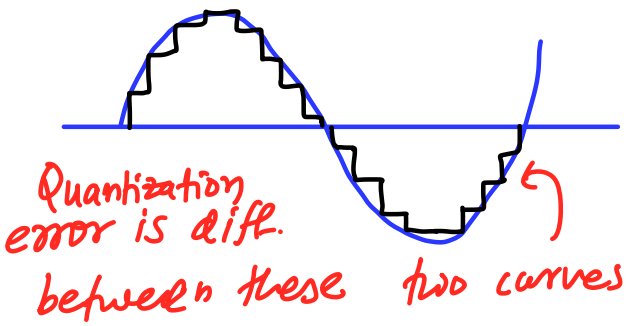
Non-ionizing - MRI, Ultrasound, Thermal imaging, OCT

ii) projected Xray - CT - Xray attenuation

Nuclear medicine - Tracer uptake
MRI - water (proton) density
ultrasound - acoustic impedance
Thermal imaging - IR radiation
OCT - optical reflectivity

(b) i) Dynamic range (ADC resolution)
voltage range
Sampling frequency

ii) Different between true analog signal value and ADC output



ii) * Input voltage to ADC should be large as possible without saturating

* Use amplifier with sufficient gain to spread small signals within the entire range of ADC

(c) i) $\lambda = \frac{\ln 2}{1.7}$

No. of hours kept idle. - 21 hours

$N = N_0 e^{-\lambda t}$

$300 = N_0 e^{-\frac{\ln 2}{1.7} \times 21}$

$N_0 = 1,569,376 \text{ kBq} = 1.57 \text{ GBq}.$

ii) $100 \times e^{-\frac{\ln 2}{1.7} \times t} = 20 \times e^{-\frac{\ln 2}{6} \times t}$

$\ln 100 - \frac{\ln 2}{1.7} t = \ln 20 - \frac{\ln 2}{6} t$

$\ln\left(\frac{100}{20}\right) = \ln 2 \left(\frac{1}{1.7} - \frac{1}{6}\right) t$

$t = \frac{\ln 5}{\ln 2} \left(\frac{1.7 \times 6}{6 - 1.7}\right)$

$t = 5.51 \text{ hours}$

Question 2

(a) In digital subtraction angiography, two images are acquired, the first before injection of the contrast agent, and the other post-injection.

i. Write an expression for the X-ray intensity (I_1) in the first scan in terms of I_0, μ_{tissue}, x_{tissue}, μ_{blood} and x_{vessel}, where I_0 is the incident X-ray intensity, μ_{tissue} and μ_{blood} are attenuation coefficients, x_{tissue} and x_{vessel} are the dimensions of the respective organs in the direction of X-ray propagation. [2]

ii. Write a corresponding expression for the X-ray intensity (I_2) for the second scan, after the injection of the contrast agent with an attenuation coefficient μ_{contrast} (Assume that $\mu_{\text{contrast}} >> \mu_{\text{blood}}$). [2]

iii. Show that the image signal intensity from static tissue is not removed by subtracting the two images. [3]

iv. Show that the signal from static tissue is removed by computing the quantity log(I_2)-log(I_1). [4]

(b) Computed Tomographic (CT) systems reconstruct a two-dimensional image slice from a series of one-dimensional X-ray projections.

i. Consider the unit disk given by

$$f(x,y) = \begin{cases} 1 & x^2 + y^2 \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Calculate its 2D Radon transform. [4]

ii. Describe **three** features of each of third and fourth generations of CT scanners. [6]

iii. Name **four** routine Quality Assurance tests performed on CT systems. [4]

b) (iii) * CT number accuracy test
* Uniformity test
* Spatial resolution test
* Slice thickness accuracy test.

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Question 2

a) i) $I_1 = I_0 e^{-(\mu_{tissue} x_{tissue} + \mu_{blood} x_{vessel})}$

ii) $I_2 = I_0 e^{-(\mu_{tissue} x_{tissue} + \mu_{contrast} x_{vessel})}$

iii) $I_2 - I_1 = \underbrace{I_0 e^{-(\mu_{tissue} x_{tissue})}}_{\text{signal intensity from static tissue not removed}} \left[\mu_{blood} x_{vessel} - \mu_{contrast} x_{vessel} \right]$

iv) $\ln(I_2) - \ln(I_1) = \ln\left(\frac{I_2}{I_1}\right) = \ln I_0 - (\mu_{tissue} x_{tissue} + \mu_{blood} x_{vessel})$

$$- \left[\ln I_0 - (\mu_{tissue} x_{tissue} + \mu_{contrast} x_{vessel}) \right]$$

$$= \underbrace{(\mu_{contrast} - \mu_{blood}) x_{vessel}}_{\text{only this term is remained}}$$

(b) ii) **third generation**

* Mutiple detectors cover entire beam
* no calibration during scan (detector drift present)
* Grid detector collimation present

fourth generation

* ring detector
* detector is stationary only source rotates
* detector drift not a problem

Question 3.

(a) Magnetic Resonance Imaging (MRI) has many advantages over other imaging modalities in clinical applications.

i. State **three** main limitations of MRI. [3]

ii. Briefly describe the main components of an MRI scanner. [6]

iii. Name **three** important requirements of an imaging magnet. [3]

(b) A spherical water phantom with a diameter of 21 cm was scanned using a 3T MR scanner kept at 24 ^{o}C. A z-gradient, $G_z = 0.5$ mT/cm is applied for the slice selection. The selected slice thickness is 3 mm.

i. Calculate the percentage of protons contributing to the MRI signal. [5]

ii. Calculate the range of Larmor frequencies of protons in the slab placed at the centre. [3]

iii. What is the bandwidth of the RF pulse required to excite this sample? [3]

iv. Name **two** MRI system parameters which can be tested using this phantom scan. [2]

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Question 3

(a) i) * High cost
* Long scan time
* Unable to image patients with metal implants.

ii) Magnet – Generate a high magnitude constant magnetic field

Gradient Coils – Produce a uniform variation in magnetic field

RF coil – Transmit RF pulse and receive return signal

iii) * High magnetic field strength
* Uniform magnetic field
* Temporal stability

b) i) percentage of protons $= \frac{V_{slice}}{V_{sphere}} \times 100$

$$= \frac{\pi R^2 t}{\frac{4}{3}\pi R^3} \times 100 = \frac{3}{4}\frac{t}{R} \times 100$$

$$= \frac{3}{4} \times \frac{3 \times 10^{-3}}{\frac{21}{2} \times 10^{-2}} \times 100 = 2.143\ \%$$

ii) $\Delta B = G_z t$

$= 0.5 \times 10^{-3} \times 3 \times 10^{-3}$

$= 1.5 \times 10^{-6}$

$\Delta W = \gamma \Delta B$

$= 267.52 \times 10^{6} \times 1.5 \times 10^{-6}$

$= 401.28$ rad/s

iii)

iv) Slice thickness accuracy
Gradient field strength and uniformity.

Question 4.

(a) Medical ultrasound is the second most commonly used imaging modality in clinical imaging.

i. State **three** advantages and **three** disadvantages of ultrasound imaging. [3]

ii. Briefly explain the ultrasound-tissue interaction mechanisms and their effect on an ultrasound image. [6]

iii. Assume that you need to image a deep-lying small structure which is located 20 cm deep, embedded within a homogeneous tissue with sound speed c = 1500 m/s and an attenuation coefficient of 1.0 dB/(cm MHz). The system is sensitive to at most a 100 dB signal loss. You can choose one of the three operating frequencies 1 MHz, 2 MHz or 5 MHz. Which frequency will give the best image of this structure? Justify your answer. [5]

iv. The ultrasound system used to view the structure in (a)iii. provides a frame rate of 100 Hz. Calculate the maximum number of lines that could be acquired. [3]

(b) Thermal imaging uses infra-red radiation emitted from the body to produce an image of temperature distribution.

i. Describe the principles of operation of thermal imaging. [4]

ii. Give **four** medical applications of thermal imaging. [2]

iii. State **four** advantages of thermal imaging in medical applications. [2]

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End of Paper.

Question 4

a) i) advantages

* High temporal resolution
* Portability
* Least expensive

disadvantages

* Limited field of view
* Operator dependent
* Difficult to interpret

ii) Reflection: when US waves encounter a boundary between different tissues with different acoustic impedence part of the wave gets reflected back to transducer

Amount of reflection depends on difference in acoustic impedance

reflection coefficient: $\frac{Z_2 - Z_1}{Z_2 + Z_1}$

Strong reflections produce bright echos on the image

Scattering: US waves scatter in multiple directions when they encounter irregular structures.

Speckle noise in image is due to this.

Attenuation: Decrease in intensity due to absorption

$I(d) = -\mu f d$

Cause acoustic shadowing

iii) $100 \geq 1 \times (2 \times 20) \times f$

$f \leq 2.5$ MHz

We can choose between 1 or 2 MHz

Higher frequency provide better spatial resolution. Therefore choose 2 MHz. and provide acceptable attenuation of 80 dB which is less than 100 dB

iv) Time for one pulse-echo cycle $T_{cycle} = \frac{2 \times 0.2\,m}{1500}$

$= \frac{1}{3750}$ s

Time to read one frame $= T_{frame} = \frac{1}{100}$ s

no. of lines per 1 frame $= \frac{T_{frame}}{T_{cycle}} = \frac{3750}{100} = 37.5$

$= 37$ lines per frame

b) i) blackbody radiation electromagnetic radiation from a body of a temperature above absolute zero. Likewise human body also emit infrared radiation

Infrared lens optical system focus infrared radiation onto the detector.

Microbolometer is a focal plane array detector which absorbs IR radiation and converts into a temperature reading

This is read using an ADC and processed to form a thermal image

ii) * Breast cancer screening
* Assessment of inflammatory conditions
* Monitoring complex regional pain syndrome
* Fever detection and infection control

iii) * Portable
* No ionizing radiation
* Power efficient
* Non invasive, non contact method.